



BIA

BOSTON
INSTITUTE OF
ANALYTICS



AI Training Assistant for New Employees

Agenda

- Introduction
- Problem Statement
- Project Objectives
- Scope of the Project
- System Architecture
- Methodology
- Python Code Demonstration
- Results
- Benefits and Future Enhancements
- Conclusion
- Questions & Answers

INTRODUCTION

Employee onboarding is a critical process in every organization. New employees often have questions regarding company policies, HR procedures, work guidelines, and organizational culture.

An AI Training Assistant helps new employees by providing instant responses to their queries, reducing dependency on HR personnel and improving the onboarding experience.

Problem Statement

- Organizations spend significant time and resources onboarding new employees. HR teams repeatedly answer common questions about company policies, benefits, leave rules, payroll, IT procedures, and workplace guidelines. Static training materials such as PDFs and presentations do not provide personalized or interactive learning experiences.
- As organizations grow, manual onboarding becomes difficult to scale and can lead to inconsistent knowledge transfer. There is a need for an intelligent AI-based assistant that provides instant, accurate, and personalized onboarding support while reducing HR effort and improving employee learning outcomes.

Objectives

- Automate the employee onboarding process.
- Provide 24×7 AI assistance for employee queries.
- Reduce repetitive HR support tasks.
- Generate personalized training recommendations.
- Conduct quizzes to evaluate employee understanding.
- Track employee learning progress.
- Improve employee engagement and knowledge retention.

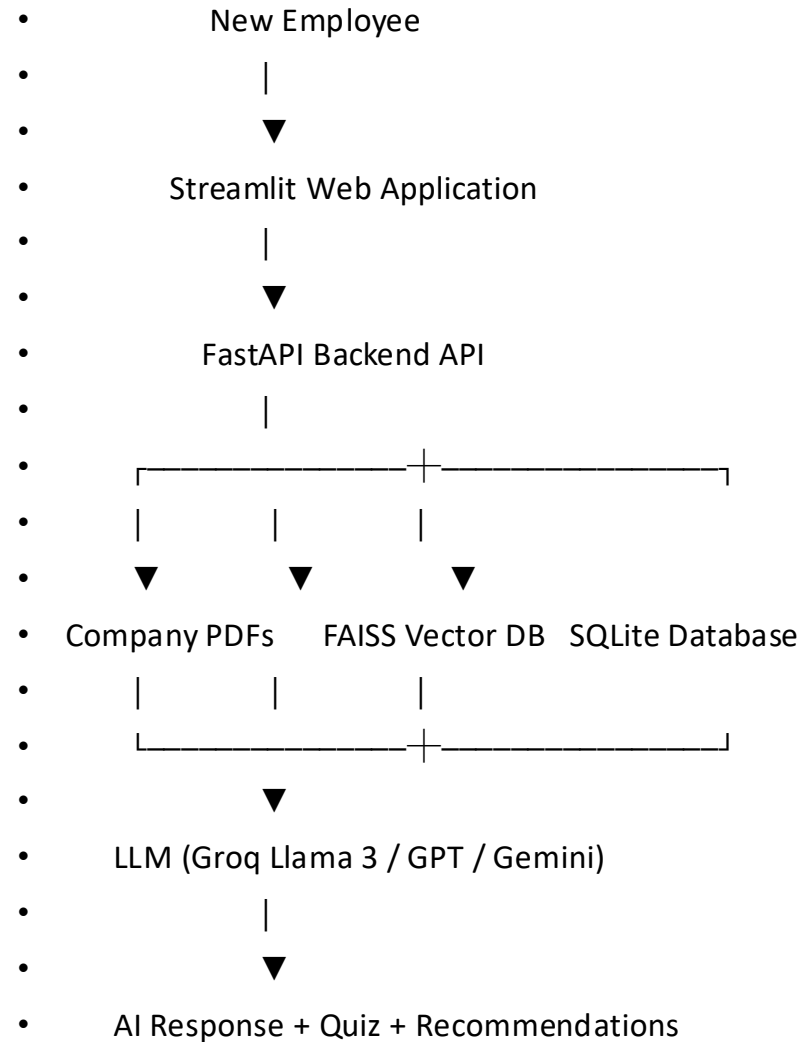
Proposed Solution

- The AI assistant enables new employees to:
- Ask questions using natural language.
- Search company policies and training documents.
- Receive accurate AI-generated responses.
- View recommended learning resources.
- Take quizzes after completing training modules.
- Monitor their learning progress.
- HR administrators can:
- Upload onboarding documents.
- Update company policies.
- View employee progress reports.
- Analyze common employee queries.

Key Features

- Employee Features
- Secure login
- AI chatbot
- Policy search
- Training recommendations
- Quiz generation
- Progress dashboard
- Frequently Asked Questions (FAQ)

System Architecture



Technology Stack

Layer	Technology
Frontend	Streamlit
Backend	FastAPI
Programming Language	Python
AI Model	Llama 3 (Groq API), GPT, or Gemini
Embeddings	Sentence Transformers
Vector Database	FAISS
Database	SQLite
PDF Processing	PyPDF2
Authentication	JWT
Version Control	GitHub

Modules

- Module 1: User Authentication
 - Employee login
 - HR/Admin login
 - Role-based access
- Module 2: Document Management
 - Upload onboarding PDFs
 - Store documents
 - Extract text
 - Create embeddings
 - Save vectors in FAISS

Module 3: AI Chat Assistant

- Employees can ask questions such as:
- How do I apply for leave?
- What are office timings?
- How do I reset my password?
- What is the reimbursement policy?
- The assistant retrieves relevant information from uploaded documents and generates an accurate response.
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Module 4: Learning Recommendation

- The assistant recommends:
- Training videos
- Company documentation
- Learning modules
- Best practices

Module 5: Quiz Generator

- Example:
- **Question:**
Which portal is used to update your emergency contact?
- A. CRM
- B. HR Portal
- C. Finance Portal
- D. Payroll Portal
- **Correct Answer:**
- B. HR Portal

Module 6: Progress Tracking

- Tracks:
- Completed lessons
- Quiz scores
- Learning history
- Weak topics
- Training completion percentage

Workflow

- Employee Login
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- Access AI Assistant
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- Ask a Question
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- Search Company Documents
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- Retrieve Relevant Content
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 - ▼
- Generate AI Response
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 - ▼
- Recommend Training Material
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 - ▼
- Generate Quiz
 - |
 - ▼
- Store Progress

Sample Questions



Employee Question	AI Response
How do I update my address?	Go to the HR Portal → Profile → Personal Information → Update Address and submit the request.
What is the leave policy?	Employees are entitled to annual, sick, and casual leave according to company policy.
Who should I contact for IT issues?	Contact the IT Help Desk using the internal support portal or email.
Where can I find the code of conduct?	The Code of Conduct is available in the Employee Handbook under the Policies section.

Advantages

- 24/7 onboarding assistance
- Faster knowledge access
- Personalized learning
- Reduced HR workload
- Interactive training
- Improved employee confidence
- Consistent information delivery
- Scalable for organizations of any size

Expected Outcomes

- Faster onboarding process
- Improved employee productivity
- Reduced repetitive HR queries
- Higher training completion rates
- Better employee engagement
- Centralized knowledge management

Future Enhancements

- Voice-enabled AI assistant
- Multilingual support
- Microsoft Teams and Slack integration
- Mobile application
- Gamification with badges and certificates
- Personalized learning paths
- HR analytics dashboard
- Calendar integration for training schedules

Conclusion

- The **AI Training Assistant for New Employee Onboarding** is a modern, intelligent solution that simplifies the onboarding process through conversational AI and document-based knowledge retrieval. By integrating LLMs with a RAG architecture, the assistant provides accurate, context-aware responses while reducing HR workload and improving employee learning. This project demonstrates practical applications of generative AI in enterprise environments and is well suited as a capstone project in Artificial Intelligence, Machine Learning, or Computer Science.

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Thank You!